

2. Maxwell's Equations and Photons

Outline Maxwell's equations and the propagation of light in vacuum and in matter

Concepts of photons and the density of states for photons

The density of states later for other (quasi-) particles, including systems of 3D, 2D, 1D, 0D.

The occupation probability of states for various groups of particles.



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2.1 Maxwell's Equations

- Maxwell's equations can be written in different ways.
- The differential form can be written as follows (bold characters symbolize vectors and normal characters scalar quantities.):

$$\nabla \cdot \mathbf{D} = \rho, \quad \nabla \cdot \mathbf{B} = 0, \quad (2.1a,b)$$

$$\nabla \times \mathbf{E} = -\dot{\mathbf{B}}, \quad \nabla \times \mathbf{H} = \mathbf{j} + \dot{\mathbf{D}}, \quad (2.1c,d)$$

$$\mathbf{D} = \varepsilon_0 \mathbf{E} + \mathbf{P}, \quad \mathbf{B} = \mu_0 \mathbf{H} + \mathbf{M}. \quad (2.1e,f)$$

- The various symbols have the following meanings:

$\mathbf{E}$ = electric field strength; $1 \text{ V/m} = 1 \text{ m kg s}^{-3} \text{ A}^{-1}$

$\mathbf{D}$ = electric displacement; $1 \text{ A s/m}^2 = 1 \text{ C/m}^2$

$\mathbf{H}$ = magnetic field strength; 1 A/m

$\mathbf{B}$ = magnetic induction or magnetic flux density; $1 \text{ V s/m}^2 = 1 \text{ T} = 1 \text{ Wb/m}^2$

ρ = charge density; $1 \text{ A s/m}^3 = 1 \text{ C/m}^3$

$\mathbf{j}$ = electrical current density; 1 A/m^2

$\mathbf{P}$ = polarization density of a medium, i.e., electric dipole moment per unit volume; 1 A s/m^2

$\mathbf{M}$ = magnetization density of the medium, i.e., magnetic dipole moment per unit volume¹; 1 V s/m^2

$\varepsilon_0 \approx 8.859 \times 10^{-12} \text{ A s/V m}$ is the permittivity of vacuum

$\mu_0 = 4\pi \times 10^{-7} \text{ V s/A m}$ is the permeability of vacuum

∇ = Nabla-operator, in Cartesian coordinates $\nabla = (\partial/\partial x, \partial/\partial y, \partial/\partial z)$
 $\dot{} = \partial/\partial t$ i.e., a dot means differentiation with respect to time.



Maxwell's Equations: Integral Form

$$\left\{ \begin{array}{l} \oiint_S \mathbf{D} \cdot d\mathbf{S} = q_0 \\ \oiint_S \mathbf{B} \cdot d\mathbf{S} = 0 \\ \oint_L \mathbf{E} \cdot d\mathbf{l} = - \iint_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{S} \\ \oint_L \mathbf{H} \cdot d\mathbf{l} = I_0 + \iint_S \frac{\partial \mathbf{D}}{\partial t} \cdot d\mathbf{S} \end{array} \right.$$



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$\nabla f(\mathbf{r}) = \text{grad } f$ The gradient of scalar function $\nabla f(\mathbf{r}) = \frac{\partial f}{\partial x} \mathbf{i} + \frac{\partial f}{\partial y} \mathbf{j} + \frac{\partial f}{\partial z} \mathbf{k}$

$\nabla \cdot \mathbf{F}(\mathbf{r})$ The divergence of vector function $\nabla \cdot \mathbf{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}$

$\nabla \times \mathbf{F}(\mathbf{r})$ The curl of vector function $\nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_x & F_y & F_z \end{vmatrix}$

$\Delta \equiv \nabla^2$. The Laplace operator Δ , if applied to a scalar field ρ :

$$\Delta \rho = \frac{\partial^2 \rho}{\partial x^2} + \frac{\partial^2 \rho}{\partial y^2} + \frac{\partial^2 \rho}{\partial z^2} \quad (2.2)$$

If applied to a vector field $\mathbf{E}$ results in:

$$\Delta \mathbf{E} = \begin{pmatrix} \frac{\partial^2 E_x}{\partial x^2} + \frac{\partial^2 E_x}{\partial y^2} + \frac{\partial^2 E_x}{\partial z^2} \\ \frac{\partial^2 E_y}{\partial x^2} + \frac{\partial^2 E_y}{\partial y^2} + \frac{\partial^2 E_y}{\partial z^2} \\ \frac{\partial^2 E_z}{\partial x^2} + \frac{\partial^2 E_z}{\partial y^2} + \frac{\partial^2 E_z}{\partial z^2} \end{pmatrix}. \quad (2.3)$$

The operators ∇ and Δ have other forms if in other than Cartesian coordinates (Polar or cylindrical coordinates)



The meanings of Maxwell's Equations

$$\nabla \cdot \mathbf{D} = \rho, \quad \nabla \cdot \mathbf{B} = 0, \quad (2.1a,b)$$

$$\nabla \times \mathbf{E} = -\dot{\mathbf{B}}, \quad \nabla \times \mathbf{H} = \mathbf{j} + \dot{\mathbf{D}}, \quad (2.1c,d)$$

$$\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P}, \quad \mathbf{B} = \mu_0 \mathbf{H} + \mathbf{M}. \quad (2.1e,f)$$

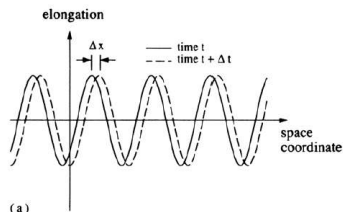
- Equations (2.1a,b) show that free electric charges ρ are the sources of the electric displacement and that the magnetic induction is source-free
- Equations (2.1c,d) demonstrate how temporally varying magnetic and electric fields generate each other. In addition, the H field can be created by a macroscopic current density $\mathbf{j}$.
- Equations (2.1e,f) are the material equations in their general form. From them we learn that the electric displacement is given by the sum of electric field and polarization, while the magnetic flux density is given by the sum of magnetic field and magnetization.



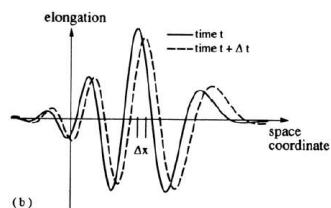
Phase velocity and group velocity

$$v_{ph} = \frac{\omega}{k} \quad v_g = \frac{\partial \omega}{\partial k} = \text{grad}_k \omega$$

For special case in vacuum, we find : $v_{ph} = v_g = c = (\mu_0 \epsilon_0)^{-\frac{1}{2}}$



A monochromatic wave



A wave packet with middle frequency ω and covering a small frequency interval $\Delta\omega$

Fig. 2.1. A harmonic wave (a) and a wave packet (b) shown at two different times t and $t + \Delta t$ to illustrate the concepts of phase and group velocity, respectively



2.2 Electromagnetic Radiation in Vacuum

- In vacuum the following conditions are fulfilled

$$\mathbf{P} = 0; \mathbf{M} = 0; \rho = 0; \mathbf{j} = 0. \quad (2.6)$$

- With the help of (2.1e,f) this simplifies (2.1c,d) to

$$\nabla \times \mathbf{E} = -\mu_0 \dot{\mathbf{H}} \quad \text{and} \quad \nabla \times \mathbf{H} = \epsilon_0 \dot{\mathbf{E}} \quad (2.7a,b)$$

- Applying $\nabla \times$ to (2.7a) and $\partial/\partial t$ to (2.7b) yields

$$\nabla \times (\nabla \times \mathbf{E}) = -\mu_0 \nabla \times \dot{\mathbf{H}} \quad \nabla \times \dot{\mathbf{H}} = \epsilon_0 \ddot{\mathbf{E}} \quad (2.8)$$

- With the properties: $\nabla \times (\nabla \times \mathbf{F}) = \nabla(\nabla \cdot \mathbf{F}) - \nabla^2 \mathbf{F}$, we find:

$$-\mu_0 \epsilon_0 \ddot{\mathbf{E}} = \nabla \times (\nabla \times \mathbf{E}) = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E} \quad (2.9)$$

$$\nabla \cdot \mathbf{E} = 0 \quad (2.10)$$

- So we get: $\nabla^2 \mathbf{E} - \mu_0 \epsilon_0 \ddot{\mathbf{E}} = 0 \quad (2.11)$

- For the most simple ones, namely on plane waves $\mathbf{E}(\mathbf{r}, t) = \mathbf{E}_0 \exp[i(\mathbf{k} \cdot \mathbf{r} - \omega t)] \quad (2.14)$

- $\mathbf{E}_0$ is the amplitude, $\mathbf{k}$ is the wave vector and ω is the angular frequency.

$$k = |\mathbf{k}| = \frac{2\pi}{\lambda_v} \quad \frac{\omega}{k} = \left(\frac{1}{\mu_0 \epsilon_0} \right)^{\frac{1}{2}} = c$$

$$\text{Helmholtz Equation} \quad (\nabla^2 + k^2) \mathbf{E}(\mathbf{r}) = 0$$



- Inserting (2.14) into (2.10) gives

$$\nabla \cdot \mathbf{E} = i\mathbf{k} \cdot \mathbf{E}_0 \exp[i(\mathbf{k} \cdot \mathbf{r} - \omega t)] = 0 \quad (2.17)$$

- This means that $\mathbf{E}_0 \perp \mathbf{k}$

$$\mathbf{H}_0 = (\omega \mu_0)^{-1} \mathbf{k} \times \mathbf{E}_0 \quad (2.18)$$

- From (2.7) we have for plane waves: $\mathbf{H} = (\omega \mu_0)^{-1} \mathbf{k} \times \mathbf{E} = \mathbf{H}_0 \exp[i(\mathbf{k} \cdot \mathbf{r} - \omega t)] \quad (2.19)$

- Furthermore we have with (2.1e,f) and (2.4):

$$\mathbf{D} = \mathbf{D}_0 \exp[i(\mathbf{k} \cdot \mathbf{r} - \omega t)] = \epsilon_0 \mathbf{E}_0 \exp[i(\mathbf{k} \cdot \mathbf{r} - \omega t)]$$

$$\mathbf{B} = \mathbf{B}_0 \exp[i(\mathbf{k} \cdot \mathbf{r} - \omega t)] = \omega^{-1} \mathbf{k} \times \mathbf{E}_0 \exp[i(\mathbf{k} \cdot \mathbf{r} - \omega t)]$$

- So we have : $\mathbf{D} \perp \mathbf{k}, \quad \mathbf{k} \perp \mathbf{B}, \quad \mathbf{B} \perp \mathbf{D}$

- In vacuum and isotropic media one has:

$$\mathbf{E}/\mathbf{D} \quad \text{and} \quad \mathbf{H}/\mathbf{B}$$

- The momentum density of the electromagnetic field is given by

$$\mathbf{\Pi} = \mathbf{D} \times \mathbf{B} \quad \mathbf{\Pi}/\mathbf{k}$$

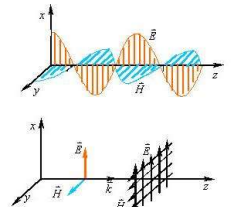
- The energy flux density, Poynting vector : $\mathbf{S} = \mathbf{E} \times \mathbf{H}$

- In vacuum and isotropic materials: $\mathbf{S}/\mathbf{\Pi}$

- The average value of Poynting vector is usually called the intensity I

- For the plane monochromatic waves, we obtain:

$$\langle \mathbf{S} \rangle = \frac{1}{2} |\mathbf{E} \times \mathbf{H}| = \frac{1}{2} \frac{1}{c \mu_0} \mathbf{E}_0^2 = \frac{1}{2} \frac{c}{\mu_0} \mathbf{B}_0^2 = \frac{1}{2} c \mu_0 \mathbf{H}_0^2$$



1) 沿z轴方向传播的基模高斯光束

$$u_{00}(x, y, z) = c_{00} \frac{\omega_0}{\omega(z)} \exp\left[-\frac{x^2 + y^2}{\omega^2(z)}\right] \exp\left\{-i\left[k\left(z + \frac{x^2 + y^2}{2R(z)}\right) - \arctg \frac{z}{f}\right]\right\}$$

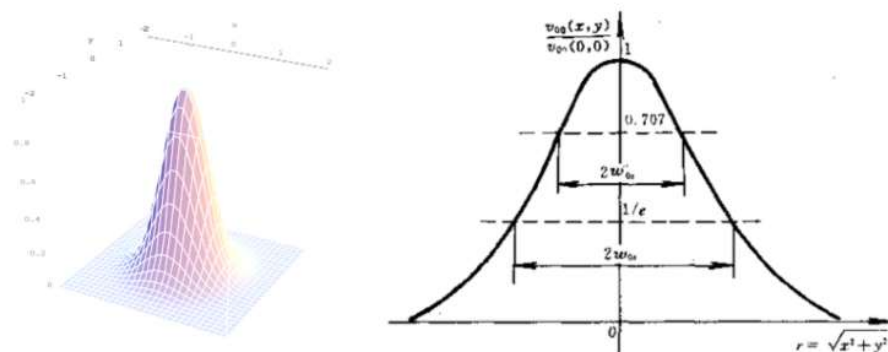
其中, c 为常数, $k=2\pi/\lambda$, ω_0 为基模高斯光束的腰斑半径, f 为高斯光束的共焦参数。曲率中心不断变化的球面波

$$\omega_0 = \sqrt{\frac{f\lambda}{\pi}}, \quad \omega(z) = \omega_0 \sqrt{1 + \left(\frac{z}{f}\right)^2}$$

$$|R(z)| = \left| f \left(\frac{z}{f} + \frac{f}{z} \right) \right|$$

对称共焦腔镜面上 $|z|=f$: $\omega_s = \omega(z) = \sqrt{2}\omega_0$

◆ 振幅分布、光斑半径

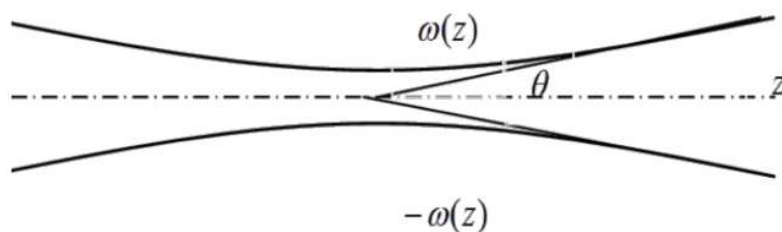


$$A(x, y, z) = A_0 \exp\left(-\frac{x^2 + y^2}{\omega(z)^2}\right), \quad I(x, y, z) = I_0 \exp\left(-\frac{2(x^2 + y^2)}{\omega(z)^2}\right)$$

2) 高斯光束在自由空间的传输规律:

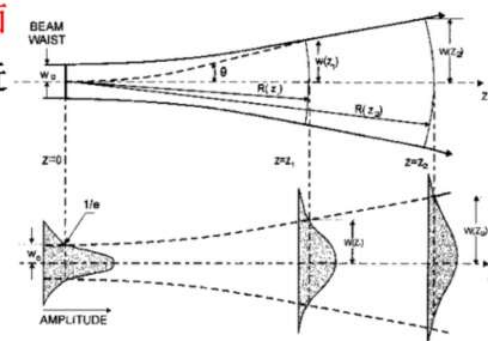
$$\omega(z) = \omega_0 \sqrt{1 + \left(\frac{z}{f}\right)^2}, \quad \theta = \lim_{z \rightarrow \infty} \frac{2\omega(z)}{z} = 2\sqrt{\frac{\lambda}{f\pi}}$$

$\pm \omega(z)$ 的渐近线夹角 θ 定义为光束的发散角



高斯光束在其传输轴线附近

- ◆ 可近似看作是一种非均匀球面波
- ◆ 曲率中心随着传输过程而不断改变
- ◆ 振幅和强度在横截面内始终保持高斯分布特性
- ◆ 等相位面始终保持为球面
- ◆ 强度集中在轴线及其附近



贝塞尔光束

Bessel 光束, 于1987年由美国J. Durnin教授首次提出, 并用环缝等装置在实验上得到了近似Bessel光束。这一发现引起了国内外学者对Bessel光束的极大兴趣, 致使近几十年来Bessel光束的相关研究和应用得到迅速发展。Bessel光束具有传播方向上不发散、中心光斑极小(仅几十微米)、在传播过程中遇到障碍物后能够自愈等特性, 在粒子操控、光学引导以及最近提出的光学拉力等方面具有广泛应用。

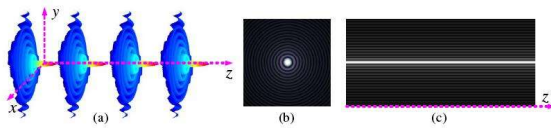
- **1983 年** Brittingham 等基于无限孔径的理想条件, 求解出麦克斯韦电磁方程的一簇新解
- **1985 年** Ziolkowski 等经过推导得到了自由空间波动方程的精确解, 并随后在水中以声波形式, 演示了近似无衍射光束的传播过程
- **1987 年** Durnin 等基于有限孔径的实验条件, 由波动方程对时间和空间分离变量求解 Helmholtz 方程时, 提出了关于自由空间标量波动方程在圆柱坐标系下的一组严格解



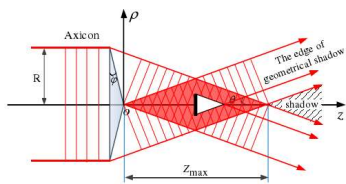
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● Bessel光束的特性

● 无衍射特性



● 自重构特性



Bessel光束自重构特性的实现需保证如下两点:
(1) 遮挡障碍物须为非透明物质;
(2) 障碍物其线度需小于Bessel光束其束腰直径。

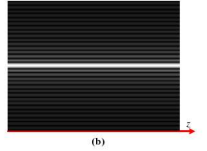
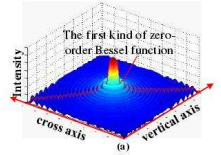


● Bessel光束产生原理

- 电场分布 E 遵循标量亥姆霍兹 (Helmholtz) 波动方程: $\left(\nabla^2 - \frac{1}{c^2} \cdot \frac{\partial^2}{\partial t^2}\right) \cdot E(r, \theta, z, t) = 0$

c——光速, t——时间函数, r, θ, z——径向、角向和轴向坐标, E——场函数

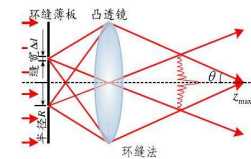
- 将 $E(r, \theta, z, t) = U(r) \cdot T(t)$ 代入上式中: $\frac{1}{c^2} \cdot \frac{T''(t)}{T(t)} = \frac{\nabla^2 U(r)}{U(r)}$
- 化简得到: $T''(t) + k^2 c^2 \cdot T(t) = 0$
 $\nabla^2 U(r) + k^2 \cdot U(r) = 0$
- 将 $\nabla^2 U(r) + k^2 \cdot U(r) = 0$ 解析为柱坐标形式: $\frac{1}{\rho} \cdot \frac{\partial}{\partial \rho} \left(\rho \frac{\partial U}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 U}{\partial \phi^2} + \frac{\partial^2 U}{\partial z^2} + k^2 U = 0$
- 采用分离变量法, 在 $z \geq 0$ 的无源自由空间场中解得: $U(r) = \exp(i\beta z) \cdot J_0(\alpha \rho)$
 $\alpha = k_r = k \sin \theta$ ——光束的径向波矢量; $\beta = k_z = k \cos \theta$ ——光束的轴向波矢量
- 化简最终得到波动方程的解为: $E(r, t) = \exp[i(\beta z - \omega t)] J_0(\alpha \rho)$
- Bessel光束的光强能量密度分布为: $I(r, t) = |\exp[i(\beta z - \omega t)] J_0(\alpha \rho)|^2 \approx |J_0(\alpha \rho)|^2$



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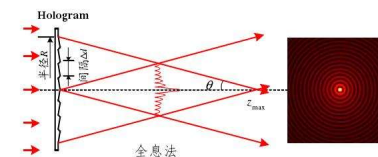
● Bessel光束的产生方式

● 环缝-透镜法



优点: 结构简单
易于实现
缺点: 利用率较低
转化效率不高

● 全息法



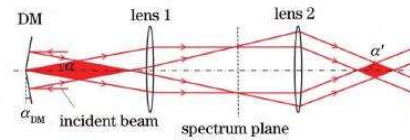
优点: 生成方式灵活
转换效率高

缺点: 全息光学器件制造困难、造价高



● Bessel光束的主要产生方式

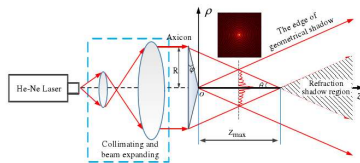
● 变形镜法



优点：校正光学系统中的像差；
承受高功率激光
缺点：技术不够成熟；
应用较少

● 轴锥镜法

McLeod 等人于 1954 年提出，是一种侧面呈圆锥形绕轴对称旋转且具有线聚焦特性的非球面透镜



优点：系统搭建简单；
转换效率较高；
缺点：表面形貌及顶点加工有一定质量要求

- ◆ Actually, all matter has some diamagnetism. And this is a rather small effect of the order of 10^{-6} , which have negligible influence on the optical properties.
- ◆ The current is driven by the electric field: $\mathbf{j} = \sigma \mathbf{E}$, and σ is small for intrinsic or weakly doped semiconductors. So $|\mathbf{j}| = |\sigma \mathbf{E}| \ll |\dot{\mathbf{p}}|$.
- ◆ We neglect $\mathbf{j}$ for weakly doped semiconductors, and get :

$$\nabla^2 \mathbf{E} - \mu_0 \epsilon_0 \ddot{\mathbf{E}} = \mu_0 \ddot{\mathbf{P}} \quad (2.26)$$

Equation (2.26) states the well-known fact that every dipole $\mathbf{p}$ and polarization $\mathbf{P}$ with a non vanishing second derivative in time radiates an electromagnetic wave.



The relationships between $\mathbf{D}$, $\mathbf{E}$ and $\mathbf{P}$

- ◆ Now we make a very important assumption.

The quantities ϵ and χ are called the dielectric function and the susceptibility, respectively.

1. Both of them are tensors, But for simplicity, can be treated as scalar quantities.
2. When frequency is dominant, we can drop $\mathbf{k}$
3. $\epsilon(0)$ is usually called the dielectric constant

$$\epsilon = \epsilon(\omega, \mathbf{k})$$

$$\chi = \chi(\omega, \mathbf{k}) = \epsilon(\omega, \mathbf{k}) - 1$$

$$\epsilon(\omega, \mathbf{k}) = \epsilon_1(\omega, \mathbf{k}) + i\epsilon_2(\omega, \mathbf{k})$$

$$\begin{aligned} \mathbf{P} &= \epsilon_0 \chi \mathbf{E} \\ \mathbf{D} &= \epsilon_0 (1 + \chi) \mathbf{E} = \epsilon \epsilon_0 \mathbf{E} \\ \epsilon &= (1 + \chi) \end{aligned} \quad (2.27)$$

2.3 Electromagnetic Radiation in Matter- Linear Optics

- Maxwell equations in matter:

$$\nabla \cdot \mathbf{D} = \rho, \quad \nabla \cdot \mathbf{B} = 0, \quad (2.1a,b)$$

$$\nabla \times \mathbf{E} = -\dot{\mathbf{B}}, \quad \nabla \times \mathbf{H} = \mathbf{j} + \dot{\mathbf{D}}, \quad (2.1c,d)$$

$$\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P}, \quad \mathbf{B} = \mu_0 \mathbf{H} + \mathbf{M}. \quad (2.1e,f)$$

$$\nabla \times \mathbf{E} = -(\mu_0 \dot{\mathbf{H}} + \dot{\mathbf{M}}) \quad \nabla \times \mathbf{H} = \mathbf{j} + (\epsilon_0 \dot{\mathbf{E}} + \dot{\mathbf{P}})$$

- There are no macroscopic free space charges i.e. $\rho = 0$ $\nabla \cdot \mathbf{E} = 0$

$$\text{Then we get : } \nabla^2 \mathbf{E} - \mu_0 \epsilon_0 \ddot{\mathbf{E}} = \mu_0 \ddot{\mathbf{P}} + \mu_0 \dot{\mathbf{j}} + \nabla \times \dot{\mathbf{M}} \quad (2.23)$$

- The inhomogeneous equation telling us that the sources of an electro-magnetic radiation field can be:

- A dipole moment $\mathbf{p}$ or a polarization $\mathbf{P}$ with a non-vanishing second-order time derivative
- A temporally varying current density
- The curl of a temporally varying magnetization

$$\nabla^2 \mathbf{E} - \mu_0 \epsilon_0 \ddot{\mathbf{E}} = \mu_0 \ddot{\mathbf{P}} \longrightarrow \nabla^2 \mathbf{E} - \mu_0(\omega) \epsilon_0 \epsilon(\omega) \ddot{\mathbf{E}} = 0$$

$$\text{If magnetic properties are included } \nabla^2 \mathbf{E} - \mu_0 \mu(\omega) \epsilon_0 \epsilon(\omega) \ddot{\mathbf{E}} = 0 \quad \text{Mostly, } \mu(\omega) \approx 1$$

$$\text{For simplicity, in the case of plane harmonic waves: } \mathbf{E}(\mathbf{r}, t) = \mathbf{E}_0 \exp[i(\mathbf{k} \cdot \mathbf{r} - \omega t)] \quad (2.32)$$

Inserting into the above, and reads:

$$\frac{c^2 k^2}{\omega^2} = \epsilon(\omega) \quad k = \frac{\omega}{c} \epsilon^{\frac{1}{2}}(\omega) = \frac{2\pi}{\lambda_v} \epsilon^{\frac{1}{2}}(\omega) = k_v \epsilon^{\frac{1}{2}}(\omega)$$

For simplicity, we introduce the complex index of refraction: $\tilde{n}(\omega) = n(\omega) + i\kappa(\omega) = \epsilon^{\frac{1}{2}}(\omega)$

So in matter, the light with k can be complex quantity :

$$k = \frac{\omega}{c} \tilde{n}(\omega) = \frac{\omega}{c} n(\omega) + i \frac{\omega}{c} \kappa(\omega) = \frac{2\pi}{\lambda_v} \tilde{n}(\omega) = k_v \tilde{n} \quad (2.36)$$

Describes the oscillatory part of the wave .

$$\text{Insertion (2.36) into (2.32): } \mathbf{E}(\mathbf{r}, t) = \mathbf{E}_0 \exp \left\{ i \left[\frac{\omega}{c} n(\omega) \mathbf{k} \cdot \mathbf{r} - \omega t \right] \right\} \exp \left[-\frac{\omega}{c} \kappa(\omega) \mathbf{k} \cdot \mathbf{r} \right] \quad (2.37)$$

$\hat{\mathbf{k}}$ is the unit vector of $\mathbf{k}$

The oscillatory spatial propagation of light in matter

A damping of light in the direction of propagation, called absorption

- In a media, the λ is connected with λ_v in vacuum: $\lambda = \lambda_v n^{-1}(\omega)$
- Compare the well-known law of absorption in the light intensity: $I(z) = I(z=0)e^{-\alpha z}$
 α is called “absorption index”
- So in the matter, the absorption $\alpha(\omega) = \frac{2\omega}{c} \kappa(\omega) = \frac{4\pi}{\lambda_v} \kappa(\omega)$
- The phase velocity of light : $v_{ph} = \frac{\omega}{\text{Re}\{k\}} = cn^{-1}(\omega)$
- The group velocity of light: $v_g = \frac{\partial \omega}{\partial k}$

Nonlinear effect

$$P(t) = \varepsilon_0 \chi^{(1)} \mathbf{E}(t) + \varepsilon_0 \chi^{(2)} \mathbf{E}^2(t) + \varepsilon_0 \chi^{(3)} \mathbf{E}^3(t) + \dots$$

- First order: linear optics
- Second order: double frequency generation, sum frequency generation.....
- Third order: self-focusing, two-photon absorption, Raman scattering....
- Higher order....



2.5 Photons and Some Aspects of Quantum Mechanics and of Dispersion Relations

$$\mathbf{E} = -\text{grad}\phi - \dot{\mathbf{A}} \quad \mathbf{B} = \nabla \times \mathbf{A} \quad (2.48)$$

ϕ is the potential, and $\mathbf{A}$ is the vector potential.

Since $\nabla \cdot (\nabla \times \mathbf{A}) \equiv 0$, so $\nabla \cdot \mathbf{B} = 0$

Because (2.48) can not exactly define the $\mathbf{A}$, for simplicity, we choose Coulomb gauge:

$$\nabla \cdot \mathbf{A} = 0$$

- The electrostatic potential ϕ obeying the Poisson equation: $\nabla^2 \phi = -\frac{\rho}{\varepsilon_0 \varepsilon(\omega)}$ (2.50)
In vacuum we still have $\rho = 0$ and the same in matter.

Second quantization:

- For simplicity, still for plane waves: $\mathbf{E}(\mathbf{r}, t) = \mathbf{E}_0 \exp[i(\mathbf{k}\mathbf{r} - \omega t)]$
So we get $\mathbf{A}(\mathbf{r}, t)$ in Fourier: $\mathbf{A}(\mathbf{r}, t) = \sum_{\mathbf{k}} \{ \alpha_{\mathbf{k},s}(t) e^{i\mathbf{k}\mathbf{r}} + \alpha_{\mathbf{k},s}^*(t) e^{-i\mathbf{k}\mathbf{r}} \} \quad \alpha_{\mathbf{k},s}(t) = \alpha_{\mathbf{k},s} e^{-i\omega_{\mathbf{k}} t}$
The classical Hamilton function: $H = E = \frac{1}{2} \int (\mathbf{E}^2 + \mathbf{H}^2) dV$
So we get: $H = 2V \sum_{\mathbf{k},s} k^2 \alpha_{\mathbf{k},s}^*(t) \alpha_{\mathbf{k},s}(t)$
 $\mathbf{k}$ is the wave vector of the plane electromagnetic and s the two possible transverse polarizations.



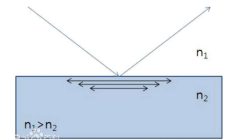
2.4 Transverse, Longitudinal and Surface Waves

- In vacuum, $\nabla \cdot \mathbf{E} = i\mathbf{E}_0 \cdot \mathbf{k} \exp[i(\mathbf{k}\mathbf{r} - \omega t)] = 0$, there is only one solution in vacuum, $\mathbf{E}_0 \perp \mathbf{k}$
- In matter, $\nabla \cdot \varepsilon_0 \varepsilon \mathbf{E} = 0$, there are two solutions, $\mathbf{E}_0 \perp \mathbf{k}$ and $\varepsilon(\omega) = 0$, we call the corresponding new solution longitudinal solutions: $\varepsilon(\omega_L) = 0 \quad \mathbf{E}_0 // \mathbf{k}$ (2.44)
- From (2.27), we see: $\mathbf{D} = 0$ and $\mathbf{E} = -\frac{1}{\varepsilon_0} \mathbf{P}$ (2.45)
- From $\nabla \times \mathbf{E} = -\dot{\mathbf{B}}$, for plane waves in nonmagnetic material: $\mathbf{H}_0 = (\omega \mu_0)^{-1} \mathbf{k} \times \mathbf{E}_0$
From (2.44), $\mathbf{H} = 0$ and $\mathbf{B} = \mu_0 \mathbf{H} = 0$ (2.47)

Longitudinal waves

- Not electromagnetic waves
- Pure polarization waves $\mathbf{E}$, $\mathbf{P}$
- $\mathbf{D}$, $\mathbf{B}$ and $\mathbf{H}$ vanishing

- Another situation: The boundary of the medium will need some extra consideration, The boundary conditions allow a surface mode, which propagates along the interface and the field amplitudes decay exponentially on both sides.



- We introduce canonically variables:

$$\alpha_{\mathbf{k},s} = \frac{i}{2k\sqrt{V}} (p_{\mathbf{k},s} - ickq_{\mathbf{k},s}) \quad \alpha_{\mathbf{k},s}^* = \frac{i}{2k\sqrt{V}} (p_{\mathbf{k},s} + ickq_{\mathbf{k},s})$$

- They fulfill the canonic equations of motion:

$$\frac{\partial H}{\partial q_{\mathbf{k},s}} = -\dot{p}_{\mathbf{k},s} \quad \frac{\partial H}{\partial p_{\mathbf{k},s}} = -\dot{q}_{\mathbf{k},s} \quad (2.51)$$

- We get: $H = \frac{1}{2} \sum_{\mathbf{k},s} (p_{\mathbf{k},s}^2 + \omega_{\mathbf{k}}^2 q_{\mathbf{k},s}^2)$, and is the usual form of the harmonic oscillator Hamilton

$$p_{\mathbf{k},s} q_{\mathbf{k},s} - q_{\mathbf{k},s} p_{\mathbf{k},s} = \frac{\hbar}{i} \quad (2.53)$$

- The electromagnetic field energy: $E_{\mathbf{k}} = \left(n_{\mathbf{k}} + \frac{1}{2}\right) \hbar \omega_{\mathbf{k}}$ with $n_{\mathbf{k}} = 0, 1, 2, \dots$ (2.54)

- Photons: $\hbar \omega_{\mathbf{k}}$
- Exchange energy with other system only in units of $\hbar \omega_{\mathbf{k}}$
- $\frac{\hbar \omega_{\mathbf{k}}}{2}$ is the zero-point energy of every mode

- We introduce operators:

$$\begin{aligned} a_{\mathbf{k},s} |n_{\mathbf{k},s}\rangle &= \sqrt{n_{\mathbf{k},s}} |n_{\mathbf{k},s}-1\rangle & \text{A new state with } n_{\mathbf{k},s}-1, \text{ annihilation operator} \\ a_{\mathbf{k},s}^{\dagger} |n_{\mathbf{k},s}\rangle &= \sqrt{n_{\mathbf{k},s}+1} |n_{\mathbf{k},s}+1\rangle & \text{A new state with } n_{\mathbf{k},s}+1, \text{ creation operator} \end{aligned}$$



- Their permutation relation is : $a_{\mathbf{k},s} a_{\mathbf{k},s}^\dagger - a_{\mathbf{k},s}^\dagger a_{\mathbf{k},s} = 1$
- Because of the operator $a_{\mathbf{k},s}^\dagger a_{\mathbf{k},s} |n_{\mathbf{k},s}\rangle = \hat{N} |n_{\mathbf{k},s}\rangle$,
Which gives the number of photons ,so called the number operator.
- Hamilton operator can be : $H = \sum_{\mathbf{k},s} \hbar \omega_{\mathbf{k},s} a_{\mathbf{k},s}^\dagger a_{\mathbf{k},s}$
- A linearly polarized wave is a coherent superposition of a left and right circularly polarized, they have a fixed-phase relation relative to each other.
- The angular momentum $S_z = S \parallel = \pm \hbar$
- Photons in thermodynamic equilibrium are described by Bose-statistics .
 $f_{BE} = \left[\exp\left(\frac{\hbar \omega}{k_B T}\right) - 1 \right]^{-1}$ T : the absolute temperature k_B : Boltzmann's constant
- For all quanta of harmonic waves, the momentum: $\mathbf{p} = \hbar \mathbf{k}$ ($\mathbf{k}$ is the real part of wave vector)

Summary:

1. Photons are bosons with spin $\pm \hbar$
2. Energy in vacuum: $E = \hbar \omega = \hbar c k$
3. Momentum: $\mathbf{p} = \hbar \mathbf{k}$

- Frequently use unit in this course:

1. The energy unit 1eV: $1\text{eV} = 1.60217733 \times 10^{-19} \text{J} \approx 1.6 \times 10^{-19} \text{J}$ $1\text{meV} = 10^{-3} \text{eV}$
2. Wave number : $1\text{cm}^{-1} = 0.123986\text{meV} \approx 0.124\text{meV}$

- Impose periodic boundary conditions

$$k_i = n'_i \frac{2\pi}{L}; n'_i = \pm 1, \pm 2, \pm 3, \dots \quad \Delta k_i = \frac{2\pi}{L}$$

- In polar coordinates in k -space, the differential volume

$$dV_k = 2dk \quad d = 1$$

$$dV_k = 2\pi k dk \quad d = 2$$

$$dV_k = 4\pi k^2 dk \quad d = 3$$

- The states number $\hat{D}(\mathbf{k})$ between k and $k+dk$:

$$\hat{D}(k)dk = g_s \frac{L}{\pi} dk \quad d = 1$$

$$\hat{D}(k)dk = g_s \frac{L^2}{2\pi} k dk \quad d = 2$$

$$\hat{D}(k)dk = g_s \frac{L^3}{2\pi^2} k^2 dk \quad d = 3$$

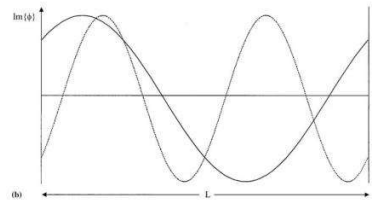


Fig. 2.3. Plane waves which have nodes at the boundaries (a) or which obey periodic boundary conditions (b)

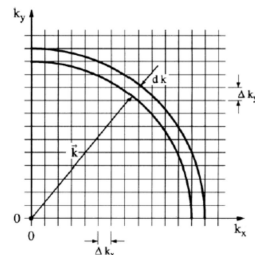


Fig. 2.4. A two-dimensional k -space in which the states are equally spaced on the k_x - and k_y -axes to derive (2.73) for $d = 2$

2.6 Density of States and Occupation Probabilities

光子, 电子, 态密度

- The density of states is a crucial concept in quantum mechanics.
- If we consider a particle which is described by a wave function $\phi(\mathbf{r})$, then the probability ω to find it in a small element of space $d\tau = dx dy dz$ around $\mathbf{r}$ is:

$$\omega(\mathbf{r}) = \phi^*(\mathbf{r})\phi(\mathbf{r})d\tau \quad (2.62a)$$

- Normalization:

$$\int_{\text{system}} \omega(\mathbf{r})d\tau = \int_{\text{system}} \phi^*(\mathbf{r})\phi(\mathbf{r})d\tau = 1 \quad (2.62b)$$

- For a plane wave, normalization requires: $\phi(\mathbf{r}) = \Omega^{-1/2} \exp(i\mathbf{k}\mathbf{r})$

$$\Omega^{-1} \int_{\text{system}} \exp(-i\mathbf{k}\mathbf{r}) \exp(i\mathbf{k}\mathbf{r}) d\tau = \Omega^{-1} \int_{\text{system}} d\tau = \Omega^{-1} V_{\text{system}} = 1 \quad (2.64)$$

- Assume: "Normalization in a box":

$$V_{\text{system}} = L^d \quad d = 3, 2, 1 \quad d = \text{dimensionality of the system} \quad \Omega^2 = L^{d/2}$$

a cube in three dimensions, a square in two, and an interval in one dimension

- The g_s considers degeneracies such as spin degeneracy, for photons $g_s = 2$
- The density of states $D(\mathbf{k})$: $D(\mathbf{k}) \propto g_s k^{d-1}$
- The density of states on the energy axis: $D(E)dE = D[k(E)] \frac{dk}{dE} dE = D[k(E)] \frac{1}{|grad_k E(\mathbf{k})|} dE$
- For photons in vacuum: $k = \frac{E}{\hbar c} = \frac{\omega}{c}$; $\frac{dk}{dE} = \frac{1}{\hbar c}$

$$\text{So: } D(E)dE = \frac{E^2}{\pi^2 (\hbar c)^3} dE \quad \text{or} \quad D(\omega)d\omega \propto \omega^2 d\omega$$

- For massive particles, described by an (effective) mass m : $E(\mathbf{k}) = \frac{\hbar^2 \mathbf{k}^2}{2m}$

$$\text{So: } D(E)dE \propto g_s E^{\frac{d}{2}-1} dE$$

- The occupation probability of the states:

Three types

1. Classical, distinguishable particles, Boltzmann statistics:

$$f_B = \exp\left[-\frac{(E - \mu)}{k_B T}\right]$$

2. For bosons, indistinguishable particles with integer spin, Bose-Einstein statistics:

$$f_{BE} = \left\{ \exp\left[\frac{(E - \mu)}{k_B T}\right] - 1 \right\}^{-1}$$

3. Fermions, or indistinguishable particles with half-integer spin, Fermi-Dirac statistics:

$$f_{FD} = \left\{ \exp\left[\frac{(E - \mu)}{k_B T}\right] + 1 \right\}^{-1}$$

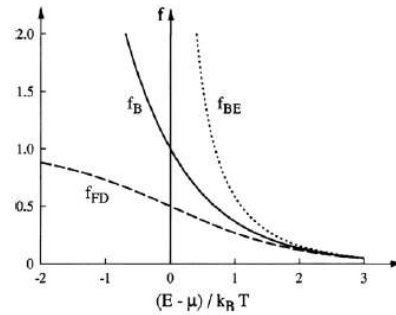


Fig. 2.5. The Boltzmann, Bose-Einstein and Fermi-Dirac distribution functions versus $(E - \mu)/k_B T$

- Example, in a three dimensional box in thermodynamic equilibrium:

$$N(\omega)d\omega = D(\omega)f_{BE}(\omega, T) \propto \omega^2 \left[\exp\left(\frac{\hbar\omega}{k_B T}\right) - 1 \right]^{-1} d\omega \rightarrow \text{Planck's Law of black-body radiation}$$

- For the energy content $U(T)$ of the radiation field:

$$U(T) = \hbar \int_0^\infty \omega N(\omega) d\omega \propto \int_0^\infty \omega^2 \left[\exp\left(\frac{\hbar\omega}{k_B T}\right) - 1 \right]^{-1} d\omega$$

$$U(T) \propto T^4 \int_0^\infty x^3 (\exp x - 1)^{-1} dx = T^4 A$$

- 玻色于1924年写了一篇推导普朗克量子辐射定律的论文，当中并没有提到任何古典物理。在开始时未能发表的挫折下，他把论文直接寄给身在德国的阿尔伯特·爱因斯坦。爱因斯坦意识到这篇论文的重要性，不但亲自把它翻译成德语，还把论文递予名望颇高的《德国物理学报》（“Zeitschrift für Physik”）发表。就是因为此次赏识，玻色能够第一次离开印度，前往欧洲并逗留两年，期间与路易·德布罗伊、居里夫人及爱因斯坦工作过。



(1894-1974)

- 爱因斯坦采取了这个概念，并把它延伸到原子去。这为预测某个现象的存在铺好了路，这个现象就是玻色-爱因斯坦凝聚，在这现象中一组高密度的玻色子（自旋为整数的粒子，以玻色命名）在超低温状态中会成为玻色-爱因斯坦凝聚体，于1995年被实验所证实。

The Nobel Prize in Physics 2001



Eric A. Cornell
Prize share: 1/3



Wolfgang Ketterle
Prize share: 1/3



Carl E. Wieman
Prize share: 1/3

The Nobel Prize in Physics 2001 was awarded jointly to Eric A. Cornell, Wolfgang Ketterle and Carl E. Wieman "for the achievement of Bose-Einstein condensation in dilute gases of alkali atoms, and for early fundamental studies of the properties of the condensates".